



5075 - Unveiling the Morphological Evolution of Galaxies in Protoclusters: Insights from JWST Imaging

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Galaxy clusters at $z < 1.5$ are found to facilitate more rapid galaxy evolution with evidence of more massive, elliptical, and quenched galaxies compared to field galaxies at similar redshift ("coeval"). It is natural to study whether such an effect is already in place in high-redshift clusters or

"protoclusters". Thus, we propose to investigate the morphology of galaxies in six well-studied protoclusters at $2.5 < z < 4.5$ from the C3VO survey that are covered by JWST NIRCам imaging from COSMOS-Web, PRIMER, JADES, and FRESCO surveys. Galaxy morphology reveals important formation processes that only structural analyses can provide, such as galaxy sizes, surface brightness profiles, merging activities, and clumpiness. This morphological study is only possible with JWST as it provides high-resolution rest-frame optical/NIR imaging for galaxies at $z > 2$, where the contribution from old stellar populations is more significant and less affected by dust. Leveraging a large number of spectra from the C3VO survey and other spectroscopy surveys, we robustly identified protoclusters/sub-structures and have spectroscopically confirmed 355 protocluster members in the selected protocluster sample. This large statistical sample of protocluster members over a large, critical redshift range allows us to (1) constrain the size evolution of protocluster members to determine how the environment affects the evolutionary track of galaxies, (2) classify morphology and characterize the evolution of the morphology-density relation, and (3) investigate the morphology of star-forming galaxies and AGN to constrain the physical processes that trigger the starbursts and AGN activities in protoclusters.

OBSERVING DESCRIPTION